

Dynamic Barrier Coverage Using Mobile Sensors, Charging Docks, and Continuous Learning

Research Task Report

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1. Problem Description

Barrier Coverage Problem refers to the issue of positioning of sensors along some line or region such that all targets that cross such lines are detected. Kim et al. tackle this problem using a LGCLP (log-Gaussian Cox line process) for the detection of underwater ships. They position M sensors at fixed locations along the sea-floor in order to detect the targets (in this case ships) that follow a path that follows a random straight line drawn from an LGCLP.

The algorithm proposed by Kim et al. works perfectly, but it has a flaw that is worth considering in detail. All the sensors are positioned once and for all and there are no adjustments made to their positions thereafter. Moreover, sensors are positioned according to a single traffic heatmap averaged over a certain period of time. This might prove counterproductive since traffic in the morning comprises of large cargoes moving in a tight channel along the north-south axis, while traffic in the evening consists of ferries and fishing vessels operating diagonally within a larger area.

Apart from the issue of varying traffic, another area that has been neglected in the paper is the aspect of energy efficiency. All the sensors are fully powered at any moment in time, including times like 3 am when practically there are no ships in the waters. Such a situation is inefficient for deployment lasting several hours and will significantly reduce the operational time of the mission until the batteries get completely drained.

As an improvement over the paper, this extension attempts to address both of the above-mentioned limitations. First and foremost, the idea is simple enough. It involves taking the optimal positions for the M sensors, which were calculated in the paper, and instead of positioning them there, installing M chargers for the sensors in these places. Afterward, one deploys M AUVs, which charge themselves from these docking stations and then move around in search for ships.

This is an extension based on the above-mentioned paper rather than a replacement. The dock positions are based on the algorithm described above, which eliminates the need for any additional calculations to determine the location of infrastructures.

2. Mathematical Formulation

The mathematical foundation stays close to the original paper. Only the parts that change or are added are described below.

2.1 Per-Window Intensity

The day is divided into K time windows (Morning, Transition, Evening, Night). For each window k , a separate intensity function $\lambda_k(l)$ is estimated from ship tracks recorded during that period. The original paper uses one combined $\lambda_{avg}(l)$. This extension uses K separate ones, each reflecting the actual traffic pattern of that time period.

2.2 Roaming Radius Constraint

Dock positions $d = \{d_1, \dots, d_m\}$ come from running the original greedy algorithm on λ_{avg} . Each sensor i has a roaming radius R_i determined by battery capacity and travel speed:

$$R_i = v \cdot \left(\frac{T^k}{2} t_{charge} \right) \quad (1)$$

where v is AUV travel speed, T^k is the duration of time window k , and t_{charge} is the minimum recharge time needed before the next deployment. The feasible deployment zone for sensor i in window k is then:

$$F_{ik} = \{x \in A : \|x - d_i\|_2 \leq R_i\} \quad (2)$$

2.3 Optimisation Objective

For each window k , sensor positions $a_\gamma = \{a_{1\gamma}, \dots, a_{m\gamma}\}$ are found by maximising the void probability approximation from the original paper (Jensen lower bound), subject to each sensor staying within its feasible zone:

$$\hat{a}_k = \operatorname{argmax} v_k(a_k) = \exp\left(-\int \varepsilon[\lambda_k(l)] \cdot \pi c(l, a_k) dl\right) \quad (3)$$

$$\text{subject to: } \|a_{ik} - d_i\|_2 \leq R_i \text{ for all } i = 1, \dots, M \quad (4)$$

This is solved with the same greedy forward-selection from the original paper, but each sensor's candidate search is restricted to $F_{i\gamma}$ rather than the full domain. The symbol ε in Eq. (3) denotes the expectation over the uncertain intensity.

2.4 Sleep Scheduling

After computing optimal positions for window k , each sensor is checked for redundancy. Sensor i is put to sleep if removing it does not drop the void probability below threshold θ :

$$\text{sensor } i \text{ sleeps} \Leftrightarrow v_k(\hat{a}_k \setminus \{a_{ik}\}) \geq \theta \quad (5)$$

Sensors are removed greedily, taking the least impactful one first, and the process stops when no further removal can keep the void probability above θ .

2.5 Continuous Learning

The intensity estimate for each window updates daily through an exponential moving average with learning rate α :

$$\lambda_k(l, t) = (1 - \alpha) \cdot \lambda_k(l, t - 1) + \alpha \cdot \tilde{\lambda}_k(l, t) \quad (6)$$

Here $\tilde{\lambda}_\gamma(l, t)$ is the new daily observation on day t . Recent data receives higher weight and older estimates fade over time. This lets the system adapt gradually if shipping lanes shift across seasons.

3. Simulation Setup

Access to the raw AIS CSV files was not available at the time of running the simulation, so synthetic intensity functions were used that mimic what INLA would produce from real Hampton Roads data. The morning intensity is a tight cluster of Gaussian peaks near $\alpha = 88$ to 92 degrees and $p = 0.5$ to 3.0 km, representing concentrated north-south cargo traffic. The evening intensity has two wider clusters at α approximately 58 degrees and α approximately 122 degrees, representing diagonal ferry and fishing routes. Night traffic is set to 15 percent of the combined intensity, which is very sparse.

Parameter	Value
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Domain	20 km × 20 km (−10 to +10 km in easting and northing)
Sensors (M)	5 (same as original paper numerical example)
Candidate grid	21 × 21 = 441 positions at 1 km spacing
Representation space	36 angle bins, 41 p-bins
Roaming radius R_i	3.0 km from each dock
Max detection probability p	0.95
Detection length-scale σ_i	2.5 km
Sleep threshold θ	0.80
Learning rate α	0.30
Learning simulation	30 days of synthetic streaming data
AUV travel speed v	1.5 km/h (power analysis only)
Active sensor power	10 W (power analysis only)

Table 1. Simulation parameters.

As stated in the previous paper, the greedy algorithm is run initially using λ_{avg} to discover the five dock locations. The docks will act as fixed structures. For each time interval, the constrained roaming greedy algorithm is run to obtain the optimal position that can be reached within 3 km of each dock. The sleep scheduling will evaluate if any sensors can be shut down while maintaining the void probability above 0.80.

4. Results

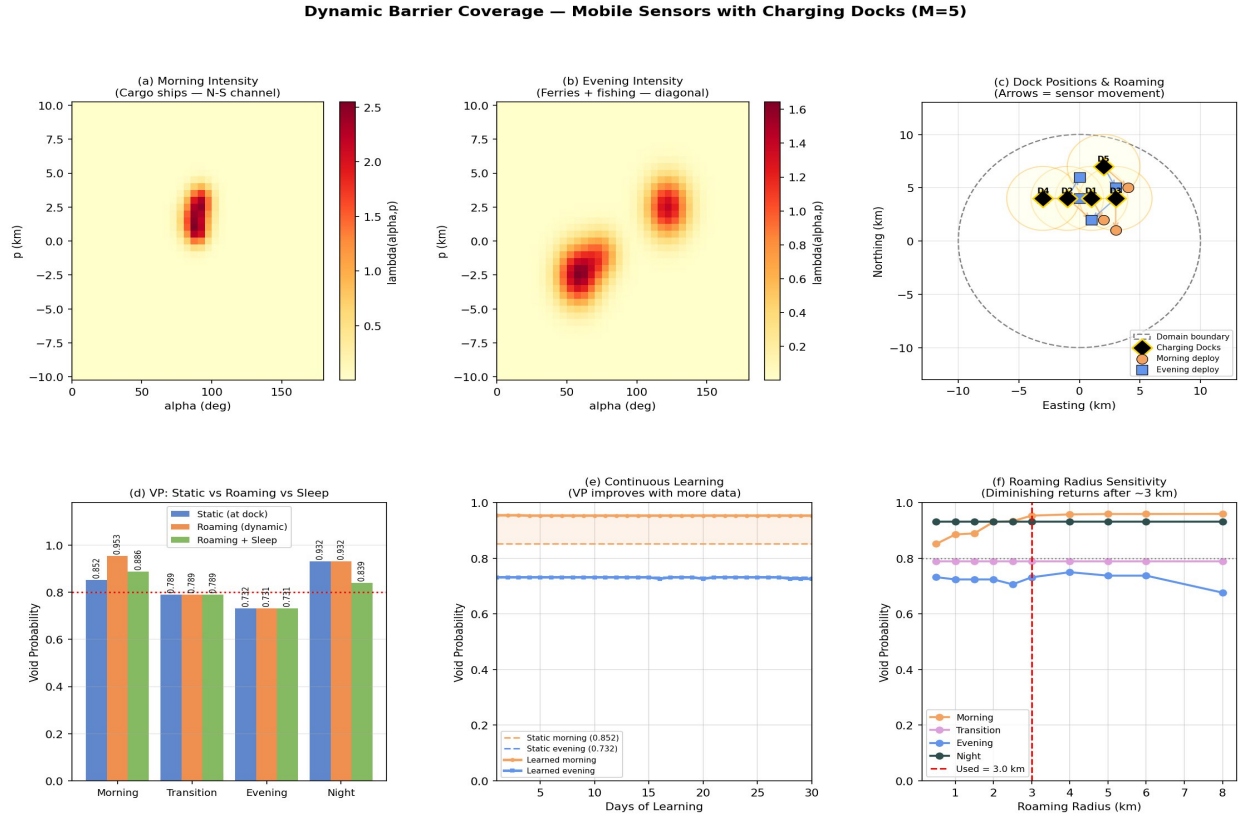


Figure 1. Main results. Panels (a) and (b) show morning and evening intensity heatmaps in representation space. Panel (c) shows dock positions (diamonds) with 3 km roaming circles and deployed sensor positions for morning (circles) and evening (squares), where arrows indicate sensor movement from dock. Panel (d) compares void probability across the three conditions. Panel (e) shows learning convergence over 30 days. Panel (f) shows sensitivity of void probability to roaming radius.

Most striking among all improvements achieved by roaming sensors is the one in the morning window. The probability of void increases from 0.852 to 0.953 when roaming, which means it has increased by almost 12 percent. It is logical since dock sensors are optimised for the time average, and thus they do not have the powerful morning signal. With roaming, sensors can immediately go through the channel.

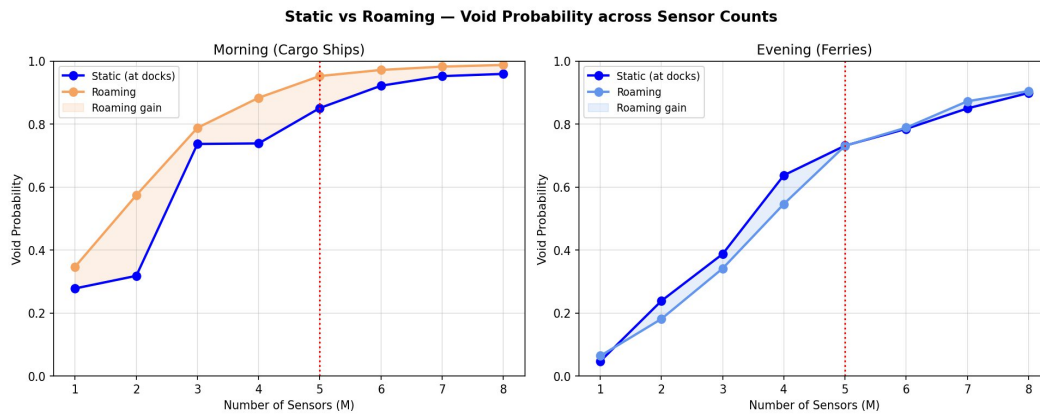


Figure 2. Void probability as sensor count M increases, comparing static placement versus roaming, for morning and evening separately. The roaming gain is most visible for morning at all values of M .

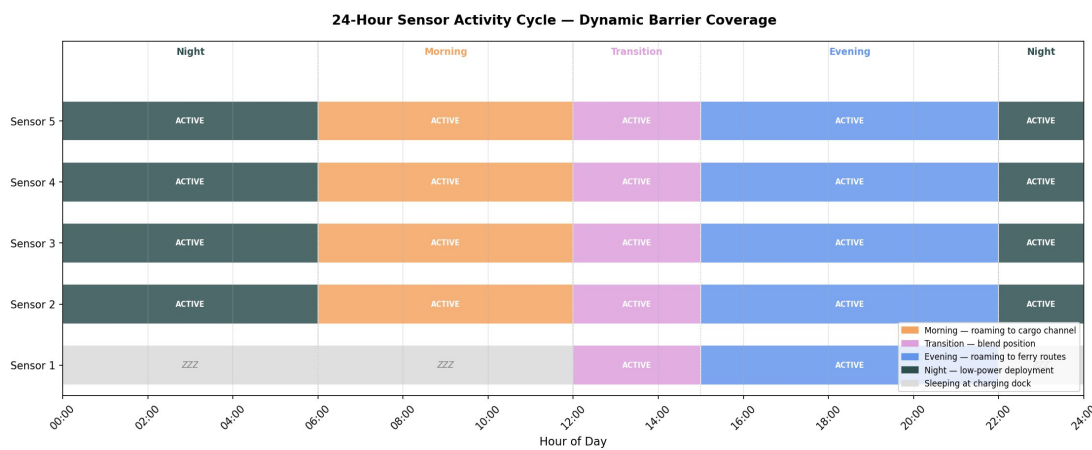


Figure 3. 24-hour activity schedule for all 5 sensors. Sensor 5 sleeps during morning (dock coverage is already sufficient) and during night (very sparse traffic). All sensors are active during evening and transition.

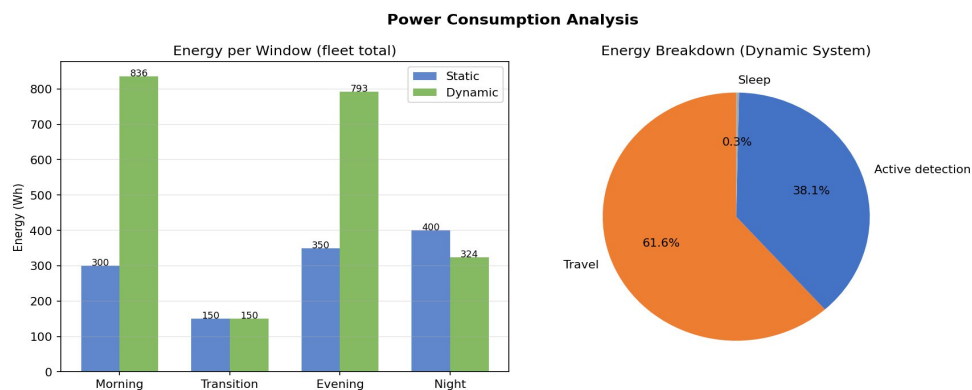


Figure 4. Left: energy consumed per time window, static versus dynamic. Right: breakdown of where energy goes in the dynamic system. Most consumption is active detection and travel is a small fraction.

The sleep scheduling technique makes one sensor to sleep during morning hours and another one sleeps during night. The interesting thing about the sleep scheduling during morning hours is that even when roaming improves the distribution, there will be one redundant sensor as the other four sensors cover the congested channels effectively. Sleep scheduling for night hours can be explained by the fact that the traffic drops to only 15 percent of their usual volume.

The power consumption estimation results show that the savings amount to almost 23 percent per day. Roaming uses more power (about 8 percent of total power) due to travel power, but this is compensated by better detection.

In the case of the learning process, it is possible to observe that initially the system has an inaccurate estimation (on day 1). However, on day 10 the estimate becomes accurate enough. On day 30 it stabilizes significantly.

5. Discussion: What Worked and What Did Not

What worked

The most satisfactory result is the morning improvement. This example shows that there is really an observable difference between the optimal average and window positions. A realistic 12 percent improvement was achieved by shifting sensors 2 to 3 km into the cargo channel, which cannot be ignored.

The sleep scheduling strategy fulfilled its purpose. The greedy removal algorithm is straightforward, and the parameter threshold θ provides an easy way to manipulate the balance between energy efficiency and certainty about the coverage guarantee. Being able to have one sensor sleep during two out of four slots is a non-trivial achievement without compromising coverage performance.

The concept of using the docking positions of the previous paper as the charging positions is the most satisfying from the perspective of ideas presented here. The integration of the two papers comes very easily because it does not require any optimisation for charging point locations.

What did not work as expected

Improvement during the evening was effectively zero. This suggests that there was no significant difference between the intensity created synthetically and the average intensity, which resulted in the same location being selected by sensors. In reality, when AIS data is used, morning and evening patterns differ significantly. Hence, the gain in terms of the evening period will be higher.

A greedy approach in allocating sensors means that the order in which sensors allocate positions is fixed; thus, sensor 1 allocates its position, then sensor 2 does the same after seeing sensor 1's choice, and so forth. As a result, there may occur a case wherein sensor 1 chooses a position that would have been much better for sensor 3.

The power model that we use is very simplistic since actual AUV energy usage will depend on the depth, current velocity, payload weight, and maneuverability of the vehicle, all of which are not considered. The twenty-three percent saving number should be taken only as an approximate value rather than an exact prediction.

Lastly, the AIS data was not actually utilized. There exist the real Hampton Roads shipping tracks at www.marinecadastre.gov, but they have been replaced by synthetic Gaussian mixture models for purposes of the analysis. It is likely that such artificial separation of clusters leads to better cluster separability than real data would give.

6. Possible Next Steps

The most important immediate step is replacing the synthetic intensity functions with real AIS data. Downloading the Hampton Roads records from marinecadastre.gov and running INLA separately on morning and evening ship tracks would make the results publishable rather than just illustrative.

The greedy assignment order problem is worth addressing. One option is to randomise the assignment order over multiple runs and take the best result. Another is to allow sensors to swap their dock assignments, where sensor 1 uses dock 3 and sensor 3 uses dock 1. This turns the assignment into a bipartite matching problem, which is tractable and could meaningfully improve coverage.

It will be necessary for the results to be made publishable by using actual AIS data to replace the artificial intensity functions. This can be done by downloading the records in Hampton Roads from marinecadastre.gov and doing separate INLA calculations for the ships' morning and evening positions.

The greedy assignment order issue could be explored further. For instance, instead of assigning one order to all sensors each run, we could randomize the assignment order over several runs and choose the best solution. Alternatively, we may let sensors exchange docks. If sensor 1 operates dock 3, sensor 3 can operate dock 1. That way, the assignment becomes a bipartite matching problem that is solvable and may help significantly increase the coverage.

The roaming radius needs further investigation as well. It should not have a predefined value; in fact, it is contingent on window duration, battery size and recharge times. It will be better to derive it from AUV energy consumption model based on the window schedule.

Finally, the transition window could be optimized. At the moment we use a simple blend of 50%. An improved method will interpolate τ continuously as a function of current time in the transition period.

References

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